

Using Thematic Mapper Data to Identify Contrasting Soil Plains and Tillage Practices

A.P. van Deventer, A.D. Ward, P.H. Gowda, and J.G. Lyon

Abstract

Landsat-5 Thematic Mapper (TM) data from 11 May 1990 for Seneca County, Ohio were used to develop TM-based probability models for classifying agricultural management practices and soil properties. Logistic regression techniques were used to relate field data for 27 fields to TM bands and indices. Field data, including tillage and drainage practices, soil plain, and soil texture, were collected from 1988 to 1990. Both soil plain and tillage logistic regression models classified 89 percent of the fields correctly. Simple ratio and normalized differences of TM bands 5 and 7 proved most useful for classifying tillage practices. TM bands 1, 2, 3, and 4 were found useful for identifying soil plain. Spectral differences were attributed to soil color differences between lake and till plain soils and surface residue differences between conservation and conventional tillage. The developed models were tested with independent data from 15 additional fields and classified 88 percent of the soil plain and 93 percent of the tillage attributes correctly. Using TM data to identify drainage practices, organic matter content, and soil texture was generally inadequate for scientific purposes.

Introduction

Scientists, engineers, and farmers have studied soil properties and agricultural management practices for many years. Knowledge of soil properties and management practices is used for purposes of planning, modeling, and environmental impact analysis of agricultural production systems. For instance, fields or watersheds might be classified to determine agricultural chemical migration into groundwater using a water quality simulation model like ADAPT (Chung *et al.*, 1992), or to determine runoff and erosion using an erosion simulation model like WEPP (Lafren *et al.*, 1991). These environmental simulation models need extensive input information regarding soil properties and management practices as well as for calibration purposes. This information can be obtained using a variety of procedures including field sampling and laboratory analysis. While many of these data collection procedures are well established and provide in-depth knowledge, they are time-consuming, labor-intensive, costly, and involve destructive sampling. A major advantage of remote sensing data acquisition techniques over conventional field and laboratory analysis is that they are non-destructive and can provide fast, low-cost information. Many researchers have studied spectral relationships of soils under laboratory conditions (Stoner *et al.*, 1980; Escadafal *et al.*, 1989) as well as field conditions (Stoner *et al.*, 1980; Coleman and Montgomery, 1987; Coleman *et al.*, 1991). Spectral characteristics

of soils under field conditions have been studied using hand-held sensors (Stoner *et al.*, 1980; Satterwaite and Henley, 1987; Coleman *et al.*, 1991), push-boom truck-mounted sensors (Clevers, 1988), and sensors aboard aircraft (Evans, 1979; Lyon *et al.*, 1986; Milton and Webb, 1987). Space-borne sensors aboard Earth-orbiting satellites have been in operation since the early 1970s and are an attractive data source as they cover a large area on a repetitive basis.

Only limited research has been conducted using satellite imagery to study soil properties and management practices (van Deventer, 1992). This is due to spatial variability, difficulties in discriminating between the many factors which influence soil reflectance, the cost of field data acquisition, and cloud cover problems. The Landsat-5 satellite thematic mapper (TM) sensor, launched in 1985, has the spatial and spectral resolution that may be useful for studying these phenomena. Hand-held TM sensors have demonstrated their usefulness (Coleman and Montgomery, 1987; Coleman *et al.*, 1991) and, as hypothesized in this study, could be useful for studies of soil properties and management practices using Landsat-5 TM satellite data. The objective of our study was to develop and empirically evaluate TM-based probability models for classifying agricultural management practices and soil properties.

Background

Surface roughness affects the spectral characteristics of soils with a decrease in roughness resulting in increased reflectance (Swain and Davis, 1978). Also, Stoner *et al.* (1980) showed for a Chalmers silty clay loam that corn residue covered fields exhibited higher reflectance than residue-free fields, especially around 1.7 μm (TM band 5). Using Landsat data, DeGloria *et al.* (1986) reported an overall classification accuracy of 81 percent for distinguishing between tilled and no-till fields. Motsch *et al.* (1990) reported an accuracy of 68 percent for Landsat TM classification of four categories of crop residue. The texture of a soil affects the spectral reflectance of soil due to its influence on the water holding capacity and because soil particle size strongly influences reflectance (Swain and Davis, 1978). Decreased reflectance occurs particularly in TM bands 5 and 7 (water absorption bands) because incoming radiation is strongly absorbed by water at these wavelengths. Weismiller and Kaminsky (1978) predicted that computer-aided analysis of satellite multispectral data would aid soil scientists in preparing soil association maps and the delineation of parent material boundaries. Roudabush *et al.* (1985) found Landsat data useful for locating soil series boundaries in preparing a detailed soil survey.

A.P. van Deventer is with Geogroep Inc., The Netherlands.

A.D. Ward and P.H. Gowda are with the Department of Agricultural Engineering and J.G. Lyon is with the Department of Civil Engineering, all at The Ohio State University, Columbus, OH 43210.

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Biswas (1987) identified Piedmont plain soils and Valley plain soils using visual interpretation of TM bands 4, 5, and 7 of Landsat satellite imagery. Using a hand-held radiometer, Coleman *et al.* (1991) found TM bands 1, 2, 3, and 7 to be key bands for identification of silt content and TM bands 2, 6, and 7 to be key bands for identification of clay content.

Presence of organic matter in a soil causes "a marked difference in reflectance throughout the 0.5- to 1.1- μ m wavelength region" (Swain and Davis, 1978). Latz *et al.* (1984) found that high organic matter content tends to cause reflectance to be low particularly in the 0.5- to 0.8- μ m region, corresponding to TM bands 1, 2, 3, and 4. Henderson *et al.* (1989) found the middle infra-red region, as well as near infra-red and visible regions, to be most important for classification of organic matter. Thompson *et al.* (1983) used multispectral scanner data from hand-held radiometer measurements to assign 246 soil series from 39 states into two categories of organic matter (i.e., 0 to 2 percent and more than 2 percent). Coleman and Montgomery (1987) and Coleman *et al.* (1991) used all radiometer bands to develop spectral relationships for organic matter. Schreier *et al.* (1988) used color aerial photographs and spectral measurements to determine rates of change and spatial distribution of organic matter in individual agricultural fields. Frazier and Cheng (1989) used Landsat TM band ratios to map organic matter levels and found TM bands 1, 3, 4, and 5 were most important.

Methods to quantify spectral differences between soil attributes focus mainly on the development of indices. Indices include original brightness values as well as transformations such as band differences, band ratios, and normalized differences of two or more bands. An important advantage of using ratios and normalized differences over original brightness values is that they help to normalize differences in solar radiation due to differences in sun angle, light intensity, and atmospheric disturbances. Vegetation Indices (VIs), in particular, have been applied successfully for crop monitoring (Perry and Lautenschlager, 1984; Thenkabail *et al.*, 1992; Thenkabail *et al.*, 1994). The most widely used VIs are the Normalized Difference Vegetation Index: i.e.,

$$NDVI = \frac{TM \text{ band } 4 - TM \text{ band } 3}{TM \text{ band } 4 + TM \text{ band } 3}, \quad (1)$$

and the Simple Vegetation Index: i.e.,

$$SVI = \frac{TM \text{ band } 4}{TM \text{ band } 3}. \quad (2)$$

In remote sensing studies of soils, only a few "Soil Indices" have been developed and applied. Satterwaite and Henley (1987) developed ratios of TM bands 1, 2, 3, and 4 from hand-held radiometer data for studies of soils in northern Nevada. Frazier and Cheng (1989) used ratios of Landsat TM bands 1, 3, 4, and 5 for organic matter content identification.

The Study Area

This study was conducted with information from farms in Seneca County, Ohio. Seneca County is in north-central Ohio and contains 140,000 hectares, with farming being the principle segment of the economy (USDA-SCS, 1980). Most areas have a slope of less than 2 percent, except for a small area in the southeast corner of the county. A mantle of Wisconsin glacial drift overlies limestone bedrock throughout most of the county. Two-thirds of the county is occupied by till plain with the dominant soils being Blount, Pandora, and Tiro. The northwestern third of the county is very flat and occupied by lake plain soils. These soils are associated with a glacial lake which formed when the Wisconsin glacier receded. Major lake plain soils are Hoytville and Nappanee.

Lake plain soils generally are darker than till plain soils and have higher clay and organic matter contents. Characteristics of the soils evaluated in this study are presented in Table 1.

Poor natural drainage is exhibited by both lake and till plain soils and adversely impacts agriculture. However, with subsurface drainage and other appropriate management practices, most soils can be highly productive for growing corn, soybeans, and wheat crops. Poor drainage and frequent saturated soil water conditions in the root zone has caused farmers to install buried subsurface drainage systems on more than 50 percent of the crop land in the county. Typically, 102-mm diameter perforated plastic pipe is installed on 10- to 20-m centers and at depths of 0.9 to 1.1 m.

Conventional tillage practices usually comprise moldboard or chisel plowing in the fall and cultivating with a disc in the spring. Climatic conditions and farmer preferences sometimes result in all tillage operations being conducted in the late winter and spring. A range of conservation tillage practices are exhibited in the region. However, common conservation tillage features are crop residue being left in the field and no moldboard or chisel plowing in most or all years.

Materials and Methods

The study was conducted with information obtained between 1988 and 1990 from commercially operated farms. Of the 36 farm fields regularly visited in 1990, nine fields were planted in winter wheat and were not considered in this study. In 1988 and 1989, one or two soil cores were obtained for 20 of the 27 fields using a Giddings hydraulic core sampler; the cores were logged by a soil scientist and analyzed for sand, silt, and clay content using standard procedures. Information on agricultural management practice was obtained from farmers through questionnaires and personal interviews. During the growing season overhead photographs (taken at a 2-m height) and soil water content samples (from the top 25 cm) were obtained every two to three weeks. Following the 1990 growing season, soil samples from the top 25 cm were obtained on 25 of the 27 fields for determination of organic matter content. All field data used in the probability model development study are reported in Table 2.

Table 3 presents the mean spectral reflectance data from a Landsat TM image of 11 May 1990 for each soil plain and tillage treatment. Conditions during the overpass were considered ideal with 0 percent cloud cover. The brightness values were analyzed using the ERDAS imaging system (ERDAS, 1991).

Tillage practices were assigned a class value of 1 (for conservation tillage) or 0 (for conventional tillage). Drainage practices were assigned a class value of 1 (for artificial subsurface drainage) or 0 (for no artificial subsurface drainage). Soil plains were assigned a class value of 1 (for lake plain soils) or 0 (for till plain soils). Similarly, corn residue was assigned a class value of 1 and soybean residue was assigned a class value of 0.

A linear logistic regression analysis was performed using the SAS computer software LOGISTIC procedure (SAS, 1990). The linear logistic equation has the form:

$$\logit(p) = \ln \left[\frac{p}{1-p} \right] = \alpha + \beta' \mathbf{X} \quad (3)$$

where p is the response probability, $\mathbf{X}$ is a vector of ordinal response variables, α is the intercept parameter, and β is a vector of slope parameters. The response probability, p expressed as a fraction, is

$$p = \frac{e^{\logit(p)}}{1 + e^{\logit(p)}}. \quad (4)$$

TABLE 1. CHARACTERISTICS OF THE SOILS EVALUATED IN THIS STUDY (USDA-SCS, 1980)

Soil Type	USDA Texture	Taxonomic Class	Surface Color ¹
Lake Plain			
Colwood	Silt Loam	Fine-loamy, mixed, mesic Typic Haplaquolls	Very dark gray (10YR 3/1)
Digby	Loam	Fine-loamy, mixed, mesic Aeris Ochraqualfs	Dark grayish brown (10YR 4/2)
Hoytville	Silty Clay Loam	Fine, illitic, mesic Mollic Ochraqualfs	Very dark gray (10YR 3/1)
Kibbie	Fine Sandy Loam	Fine-loamy, mixed, Aquollic Typic Hapludalfs	Very dark gray (10YR 3/1)
Nappanee	Silt Loam	Fine, illitic, mesic Aeris Ochraqualfs	Dark grayish brown (10YR 4/2)
Randolph	Silt Loam	Fine, mixed, mesic Aeris Ochraqualfs	Dark grayish brown (10YR 4/2)
Till Plain			
Belmore	Loam	Fine-loamy, mixed, mesic Typic Hapludalfs	Dark brown (10YR 4/3)
Blount	Silt Loam	Fine, illitic, mesic Aeris Ochraqualfs	Dark grayish brown (10YR 4/2)
Lenawee	Silty Clay Loam	Fine, mixed, nonacid, mesic Mollic Haplaquepts	Very dark grey (10YR 3/1)
Glynwood	Silt Loam	Fine, illitic, mesic Acric Hapludalfs	Dark brown (10YR 4/3)
Millgrove	Silt Loam	Fine-loamy, mixed, mesic Typic Agiaquolls	Very dark grayish brown (10YR 3/2)
Milton	Loam	Mixed mesic Typic Hapludalfs	Dark Brown (10YR 4/3)
Pandora	Silt Loam	Fine, mixed mesic Typic Ochraqualfs	Dark grayish brown (10YR 4/2)

¹All soil colors are based on wet soils. Values in parenthesis are the hue, value, and chroma as described in Soil Taxonomy (USDA-SCS, 1975).

TABLE 2. FIELD MANAGEMENT PRACTICE AND SOILS INFORMATION

Soil Plain & Field	Soil Type	Organic Matter (%)	Soil Water (g/g)	Tillage ¹	Drainage ²	Crop Residue	Sand (%)	Silt (%)	Clay (%)
Lake Plain									
90-06	Colwood	1.9	0.23	1	1	0	—	—	—
90-09	Hoytville	3.0	0.25	0	1	0	25	42	33
90-10	Hoytville	1.5	0.24	0	1	1	—	—	—
90-11	Randolph	4.1	0.31	0	1	0	40	37	23
90-16	Hoytville	1.9	0.22	1	0	1	22	40	38
90-20	Hoytville	2.7	0.27	0	1	0	25	41	34
90-22	Hoytville	1.9	0.29	1	1	1	—	—	—
90-23	Digby	1.7	0.18	0	0	0	46	37	17
90-24	Nappanee	2.0	0.21	0	0	1	18	58	24
90-25	Hoytville	2.5	0.23	1	1	1	—	—	—
90-28	Nappanee	1.6	0.24	0	0	1	18	59	23
90-33	Hoytville	3.0	0.29	1	1	0	25	43	32
Means		2.3	0.25				27.4	44.6	28.0
Till Plain									
90-01	Pandora	1.6	0.23	1	1	1	—	—	—
90-04	Blount	2.4	0.25	0	0	1	45	38	17
90-29	Belmore	1.2	0.23	0	1	1	46	38	17
90-30	Belmore	1.0	0.15	0	1	0	56	33	11
90-31	Blount	1.7	0.19	1	0	0	25	43	32
90-32	Blount	3.4	0.24	0	0	0	26	50	24
90-42	Blount	1.3	0.20	1	0	0	—	—	—
90-43	Blount	1.8	0.21	0	0	1	25	52	23
90-44	Blount	—	0.21	0	0	0	25	52	23
90-45	Blount	1.7	0.23	1	1	1	21	52	27
90-46	Blount	1.5	0.22	0	0	1	21	52	27
90-47	Pandora	1.3	0.20	1	0	1	20	53	27
90-48	Blount	1.8	0.21	1	0	0	31	51	18
90-49	Blount	—	0.19	0	1	1	—	—	—
90-55	Blount	2.2	0.25	0	1	1	26	50	24
Means		1.8	0.21				30.5	47.0	22.5

¹ 1 = conservation tillage; 0 = conventional tillage.

² 1 = no drainage; 0 = subsurface drainage.

³ 1 = soybeans; 0 = corn.

TABLE 3. MEAN BRIGHTNESS VALUES FOR EACH FIELD.¹

Tillage Practice & Statistic	TM1	TM2	TM3	TM4	TM5	TM7
Lake Plain						
Conservation Tillage						
Mean	87.6	43.6	55.0	65.8	145.2	74.0
Standard Deviation	4.8	3.4	5.9	7.8	15.4	5.2
Conventional Tillage						
Mean	83.1	40.9	50.1	57.3	115.3	64.6
Standard Deviation	2.5	2.6	4.1	1.7	4.4	6.6
Till Plain						
Conservation Tillage						
Mean	91.3	47.7	59.2	73.7	143.8	74.3
Standard Deviation	5.2	3.8	5.4	7.1	12.4	7.4
Conventional Tillage						
Mean	93.0	50.0	64.0	72.3	139.2	78.0
Standard Deviation	4.7	3.6	5.3	5.8	11.6	7.2

¹ TM1, TM2, etc. are TM bands 1, 2, etc.

The LOGISTIC procedure in SAS uses a one-step jackknife procedure to obtain new parameter estimates when classifying ordinal data. The method reduces the bias associated with estimating the error count based on the same data set that was used to develop the logistic regression equation.

It is necessary to specify a cut point response probability to classify the outcome of an event occurring. Selection of a cut point will normally depend on the application. In our effort to develop logistic models, knowledge of the actual outcome was known and the cut point which resulted in the most fields being correctly classified is reported.

TM indices were developed by using combinations of two bands selected from all seven bands and combining them into difference indices, sum indices, product indices, ratio indices, and normalized difference indices. A logistic regression analysis was performed for each class variable with (1) brightness values for each TM band; (2) each difference, sum, product, and normalized difference index; (3) a stepwise analysis where successive significant TM bands and indices were added to the logistic regression model; and (4) class variables being added to the model as explanatory variables. The analysis was performed with mean brightness values for the whole field and brightness values for a single pixel in the proximity of the field data collection activities. This approach was used because the results should apply to the whole field, but the most knowledge was available for the location in the field where data were obtained. Pixels along the boundaries of each field were omitted to minimize influences of mixed pixel effects and uncertainties in the boundaries.

A linear regression analysis was also conducted to determine if organic matter content and the percent silt, sand, and clay could be related to mean TM brightness values and indices for a field.

The most significant logistic regression models, based on single pixel and whole field data, were evaluated on 15 fields which were monitored in 1988 or 1989. Tillage information for these fields was obtained from questionnaires which farmers completed in 1988 or 1989. None of these fields were visited in 1990.

Results and Discussion

Soil Plain Analysis

As expected, most lake plain soils were darker and exhibited higher organic matter and clay contents than did the till

plain soils (Table 2). However, the correlation between the TM data, organic matter content, and percent sand, silt, and clay was insufficient for most scientific purposes. Regression equations containing four different indices were only able to account for 52 percent, 70 percent, and 51 percent of the variability in the percent sand, silt, and clay, respectively. For sand content, combinations with TM band 5, a water absorption band, were most dominant, followed by combinations with TM bands 4 and 7. For silt content, combinations with TM band 5 were most dominant, followed by combinations with TM band 2. For clay content, combinations with TM band 4 were most dominant, followed by combinations with TM band 1 and 6. Complete details of this study are reported by van Deventer (1992). The results for silt and clay partially confirmed the results reported by Coleman *et al.* (1991) in hand-held radiometer studies. Lake and till plain soils were visually different when TM bands 4, 5, and 7 were assigned to the color RGB monitor of the imaging system, confirming results reported by Biswas (1987). The most significant logistic regression models based on mean brightness values for a field are summarized in Table 4. Slightly better results were obtained with the single pixel data (not presented). All models are significant at the 99 percent level and have logit slope and intercept terms which are significant at the 95 percent level. A model using either TM band 2 or band 4 was able to correctly classify 88.9 percent of the fields.

Using more than one band in a model resulted in no improvement in the number of correctly classified fields. Use of drainage, tillage, or the type of crop residue as explanatory variables did not improve the classification capability. The probability of fields having till plain soil are reported in Table 5 for the logistic regression model based on TM band 2. Fields 90-6, 90-22, and 90-24 are ranked incorrectly by the model. The Seneca County soil survey (USDA-SCS, 1980) shows Field 90-6 consisting predominantly of lake plain Colwood soils and a small amount of Kibbie. When wet, these soils are a very dark gray (Table 1). However, on the day of the satellite overpass slides of the field show that the soil surface was a light gray. Field 90-22 consists primarily of Nappanee soils and a small amount of Hoytville. Nappanee soils are similar to till plain Blount soils (USDA-SCS, 1980) and have similar color characteristics (Table 1). A qualitative evaluation of the RGB monitor showed that Fields 90-6 and 90-22 were representative of other locations with Colwood and Nappanee soils. Field 90-24 has a Hoytville soil and the reason for the misclassification based on TM data is less apparent. Field notes and slides taken on the day of the overpass indicate that the field had recently been planted and the soil surface was a mixture of light gray dry areas and darker wet areas.

The results indicate that the main reason for the spectral differences between the lake and till plain soils was due to differences in soil color. Lake plain soils were generally gray and dark while till plain soils were brown and lighter. The

TABLE 4. SOIL PLAIN LOGISTIC REGRESSION MODEL DEVELOPMENT RESULTS.¹

TM Band	Intercept	Slope	Cut Point (%)	Correct Predictions (%)		
				All Fields	Lake Plain	Till Plain
Band 1	-30.297	0.346	50	81.5	80.0	83.0
Band 2	-24.131	0.558	32	88.9	100.0	75.0
Band 3	-15.152	0.271	32	81.5	93.3	66.7
Band 4	-17.118	0.261	56	88.9	93.3	83.3
Band 5	-7.403	0.057	46	81.5	93.3	66.7

¹ Intercept and slope terms are reported for each logit equation. For example, logit (p) = -24.131 + 0.558 TM band 2.

TABLE 5. TILL PLAIN PROBABILITY ESTIMATES FOR THE MODEL DEVELOPMENT FIELDS USING A TM BAND 2 BASED LOGISTIC MODEL.

Field (Number)	Soil Plain ¹	TM Band 2 Brightness Value	Till Plain Probability (%)
90-10	1	38	5.1
90-11	1	38	5.1
90-16	1	39	8.6
90-20	1	40	14.1
90-23	1	40	14.1
90-28	1	41	22.2
90-25	1	41	22.2
90-33	1	43	46.6
90-09	1	43	46.6
90-29	0	44	60.4
90-45	0	44	60.4
90-31	0	45	72.7
90-42	0	45	72.7
90-32	0	46	82.3
90-24	1	46	82.3
90-47	0	47	89.0
90-22	1	47	89.0
90-30	0	47	89.0
90-06	1	48	97.7
90-47	0	50	97.7
90-55	0	50	97.7

¹ 1 = lake plain; 0 = till plain.

results are consistent with the work by Cipra *et al.* (1980) who showed, with Landsat Multispectral Scanner data, that color was the most distinguishable soil property. They report spectral response differences including 3/1 (value and chroma) and 4/2 color groupings which are representative of Hoytville and Blount Soils. The ability of TM bands 1, 2, 3, and 4 to describe spectral responses of soils is in agreement with the work of Condit (1970). In a study with 160 soil samples, Condit found that soil properties could be characterized by 0.45 μm (within TM band 1), 0.54 μm (within TM band 2), 0.64 μm (within TM band 3), 0.74 μm (between TM band 3 and 4), and 0.86 μm (within TM band 4).

Tillage and Drainage Analysis

Fields with conservation tillage generally exhibited higher mean brightness values than conventionally tilled fields, particularly for lake plains soils and TM bands 4 and 5 (Table 3), confirming the results reported in hand-held radiometer studies (e.g., Stoner *et al.*, 1980). The most significant logistic regression models are summarized in Table 6. Combinations of TM band 5 with bands 1, 3, or 7 proved to be useful for tillage identification purposes. The best results were obtained with combinations of TM bands 5 and 7, namely, a Simple Tillage Index: i.e.,

TABLE 6. TILLAGE LOGISTIC REGRESSION MODEL DEVELOPMENT RESULTS.¹

Band or Index ²	Intercept	Slope	Cut Point (%)	Correct Predictions (%)		
				All Fields	Conservation Tillage	Conventional Tillage
TM5	10.215	-0.072	62	70.4	68.8	72.7
R15	-19.404	29.949	56	81.5	87.5	72.7
M15	8.785	40.947	56	81.5	87.5	81.8
D35	10.931	0.135	44	85.2	93.8	72.7
STI	45.218	-23.998	64	88.9	93.8	81.8
NDTI	30.464	-99.483	62	88.9	93.8	81.8

¹ Intercept and slope terms are reported for each logit equation. For example, logit (p) = 45.218 - 23.998 STI.

² R15 = ratio of bands 1 and 5; M15 = normalized difference of bands 1 and 5; D35 = difference between bands 3 and 5.

$$STI = \frac{TM \text{ Band } 5}{TM \text{ Band } 7}, \quad (5)$$

and a Normalized Difference Tillage Index: i.e.,

$$NDTI = \frac{TM \text{ Band } 5 - TM \text{ band } 7}{TM \text{ Band } 5 + TM \text{ Band } 7}. \quad (6)$$

Obtaining similar results with different combinations of brightness values for two bands is consistent with the results presented by Perry and Lautenschlager (1984). For example, Equation 6 can be rewritten as:

$$NDTI = \frac{STI - 1}{STI + 1}. \quad (7)$$

This functional equivalence is supported by the result that for each of our fields the ratio of NDTI to STI was almost constant and only varied between 5.8 and 6.4.

Table 7 shows the probability of a field being classified as conventional tillage based on STI. Fields 90-1 and 90-28 are ranked incorrectly by the model. Slides showed Field 90-1 containing a lot of residue and was not recently plowed or cultivated. The reason for the marginally incorrect ranking of this field as conventional tillage is not apparent. The incorrect ranking of Field 90-28 as having conservation tillage is probably because there was some residue and the soil surface had a mottled appearance.

There was no significant difference in soil water content between the conventional tillage and conservation tillage fields (0.23 g/g and 0.22 g/g in the top 25 cm, respectively). Analysis of the slides indicated that the conservation tilled fields with residue cover were visually different from the conventionally tilled fields. The main difference was the presence of decaying crop residue on the fields with conservation tillage.

TABLE 7. CONVENTIONAL TILLAGE PROBABILITY ESTIMATES FOR THE MODEL DEVELOPMENT FIELDS USING AN STI BASED LOGISTIC MODEL.¹

Field (Number)	Tillage ²	STI	Conventional Tillage Probability (%)
90-06	1	2.101	0.5
90-31	1	2.015	4.2
90-33	1	1.986	8.0
90-42	1	1.970	11.3
90-28	0	1.964	12.6
90-47	1	1.963	13.0
90-22	1	1.962	13.4
90-45	1	1.957	14.8
90-16	1	1.893	44.6
90-24	1	1.879	53.3
90-48	1	1.873	56.6
90-11	0	1.856	66.9
90-01	1	1.851	69.2
90-10	0	1.850	69.5
90-25	0	1.846	71.4
90-49	0	1.831	78.2
90-09	0	1.831	78.3
90-20	0	1.819	82.5
90-46	0	1.816	83.8
90-32	0	1.808	85.9
90-29	0	1.806	86.9
90-23	0	1.797	89.1
90-43	0	1.791	90.3
90-44	0	1.785	91.6
90-55	0	1.773	93.5
90-04	0	1.756	95.6
90-30	0	1.716	98.3

¹ Logit (p) = 45.218 - 23.998 STI.

² 1 = conservation tillage; 0 = conventional tillage.

The drainage practice was the least identifiable of all field parameters, and no significant models were developed. The drained fields were expected to have lower soil water content due to drainage of soil water from the profile. However, in this study the average soil water content in the top 25 cm was not significantly lower. In fact, the mean soil water content was slightly higher, 0.24 g/g and 0.22 g/g for the drained and undrained fields, respectively. TM bands 5 and 7, close to the water absorption bands, exhibited only small mean differences in brightness values between the drained and undrained fields (135 versus 139 and 73 versus 75, respectively). The analysis needs to be conducted with field and satellite data for conditions shortly after a drainage event when there is more variability in soil water conditions.

Model Evaluation Study

A soil plain analysis for the 15 independent fields was conducted with logistic probability models for TM band 2 and for TM band 4. These models were chosen because in the development study they correctly predicted the soil plain for the most fields (Table 4). The performance of models based on TM band 2 or TM band 4 was similar, and only Field 89-56 was ranked incorrectly. This field contains Glynwood soils which are dark brown and form in upland glacial till materials with a high lime content.

If equal weighting (cut point of 50 percent) is given to lake and till plain soils, the equation developed with the single pixel data correctly classified 13 of the 15 fields. The equation based on whole field data ranked the fields in the same order as the single pixel equation. However, based on a cut point probability of 50 percent, only 11 of the 15 fields are classified correctly.

The results for the probability model using TM band 4 correctly ranked Field 89-56 but incorrectly ranked Field 88-54 which is the only field with a Pandora soil. However, the most notable and only other difference in the results was that Field 88-12 had the highest brightness values for TM band 4 and was incorrectly ranked the least likely field to have lake plain soils. If equal weighting is given to lake and till plain soils, the equation developed with the single pixel or whole field data correctly classified 11 of the 15 fields.

The evaluation of the tillage models gave similar results for the logistic probability models which were developed with single pixel and whole pixel brightness values for either the Simple Tillage Index or the Normalized Difference Tillage Index. Only Field 89-26 was incorrectly ranked, and the reason for this cannot be determined as this field was not visited during 1990. The farmer could have switched tillage practices, left the residue in the field, or the field might have been fallow and covered in weeds.

If equal weighting is given to conventional and conservation tillage, the results for the equations developed with both single pixel and whole field data correctly classify 14 of the 15 fields. However, the single pixel equation provides a sharper transition between a low and high probability of a field having conventional tillage.

Summary and Conclusions

Using linear logistic regression, Landsat TM-based regression equations were developed for soil plain and tillage. The results confirmed previous studies which showed that brightness values for wavelengths corresponding to TM bands 1, 2, 3, and 4 are particularly useful for identifying soil properties. Data from the field investigations also showed that color was the most distinguishable soil property. The most important result was that TM data could be used to distinguish between conventional and conservation tillage even in the presence of contrasting soils. Particularly useful indices for identifying tillage differences were the functionally equivalent Simple

Tillage Index (STI) and Normalized Difference Tillage Index (NDTI). Functional equivalence of indices like STI and NDTI is consistent with the work of Perry and Lautenschlager (1984).

Identification of tillage practices using satellite imagery could be helpful for erosion and hydrologic modeling. An erosion simulation model like WEPP (Lafren *et al.*, 1991) could be provided with remotely sensed tillage information for purposes of model input, analysis, and calibration. Also, current European Economic Community (EEC) legislation requires monitoring of areas where farmers receive set-aside subsidies (Jürgens, 1993). Remotely sensed tillage information could potentially assist EEC governments to better monitor the set-aside areas. Automated classification of tillage practices should be more cost effective and faster over large areas such as the EEC, compared to manual interpretation of images, which DeGloria *et al.* (1986) found to be more cost effective for smaller areas.

Similarly, the encouraging results for soil plain identification using remotely sensed data could be useful for government agencies, engineers, and farmers who are trying to better plan, model, and analyze agricultural production systems. Logistic equations relating drainage practices to TM data were least promising due to lack of soil water content differences between drained and undrained fields on the overpass day.

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